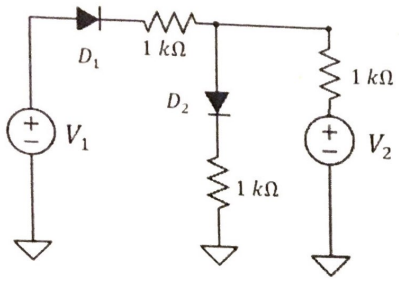
Name KEY

ID

Mark /100

α=**Sum** of the indicated digits of your ID number

1) For the circuit below, find the diode currents for each case and record in the table. Use the ideal diode model. Show all steps.



	V_1	V_2	I_{D1}	I_{D2}
(1)	α V	-5 V		
(2)	-5 V	α V		

(1) D_1 is ON for any value of α.

D_2 is OFF for $\alpha < 5$

$\Rightarrow I_{D1} = \frac{\alpha+5}{2k}$; $I_{D2} = 0$

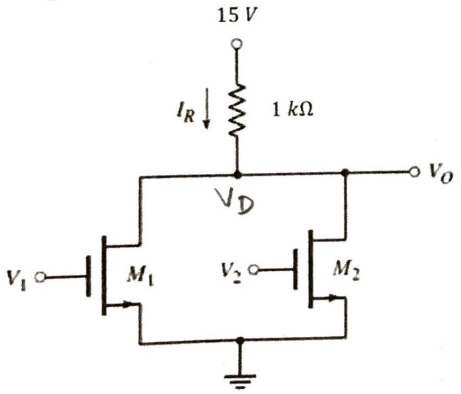
D_2 is ON for $\alpha \geq 5$

$\Rightarrow I_{D2} = \frac{\alpha-5}{3k}$; $I_{D1} = \alpha - I_{D2}$

(2) D_1 is always OFF

$I_{D1} = 0$; $I_{D2} = \alpha/2k$

2) In the MOSFET DC circuit given below, the transistors are identical with V_t =1 V, k=1 mA/V^2, and λ=0. Find the indicated quantities for each case and record in the table. Show all steps.



	V_1	V_2	I_R	V_O
(1)	α V	0 V		
(2)	α V	α V		

(1) M_2 is cutoff ($V_{GS} < V_t$)

$I_R = I_1$

Assume Saturation.

$I_1 = k(\alpha-1)^2$

$V_D = 15 - I_1$

Check mode

V_D ? $\alpha - 1$

$\Rightarrow$ $\alpha \leq 4$ Sat

$\alpha > 4$ Triode

(2) Both M_1 & M_2 ON

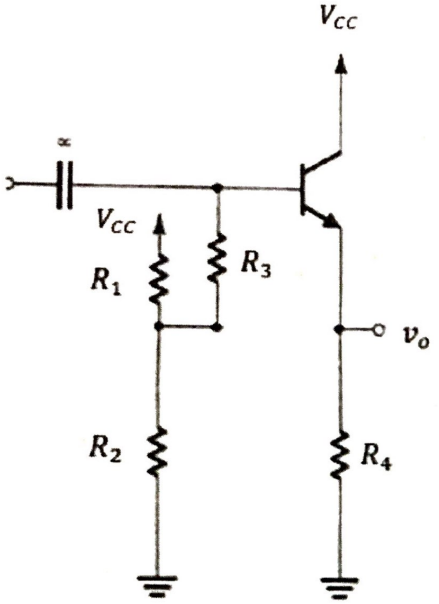
$I_R = 2I$

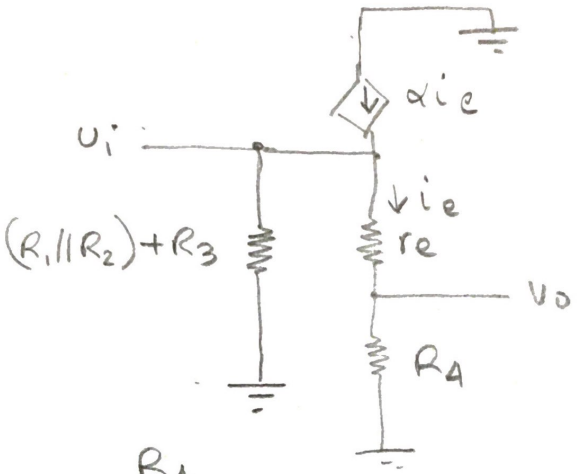
$V_D = 15 - 2I$ $\Rightarrow$ $\alpha \leq 3$ Sat

$\alpha > 3$ Triode

3) Consider the amplifier circuit below. The transistor is biased at g_m=50 mA/V and has β=100.

Ignoring r_o, find values for R_1 to R_4 such that the amplifier <u>current</u> gain equals to α (A/A).





$$\frac{v_o}{v_i} = \frac{R_4}{R_4 + r_e}$$

$$R_i = \left[(R_1 \| R_2) + R_3\right] \| (\beta+1)(r_e + R_4)$$

$$A_i = A_v \frac{R_i}{R_4} = \alpha$$

$r_e = 19.8\ \Omega \Rightarrow$ choose R_4 for example and find values for $R_1 - R_3$ to satisfy A_i.

MOSFET Current Equations	Triode	$I_D = k\,[2(V_{GS} - V_t)V_{DS} - V_{DS}^2]$	Saturation	$I_D = k\,(V_{GS} - V_t)^2(1 + \lambda V_{DS})$
BJT Current Equations	Active	$I_C = \beta I_B, \quad I_E = (\beta + 1)I_B$	Relations	$I_E = I_C + I_B$
Diode Current Equations	Forward	$I_D = I_s\,(e^{V_D/nV_T} - 1)$	Roots	$x = (-b \pm \sqrt{b^2 - 4ac})/2a$
Amplifier Equations	Gain	$A_v = v_o/v_i, \quad A_i = i_o/i_i$	Resistance	$R_i = v_i/i_i, \quad R_o = v_o/i_o$